

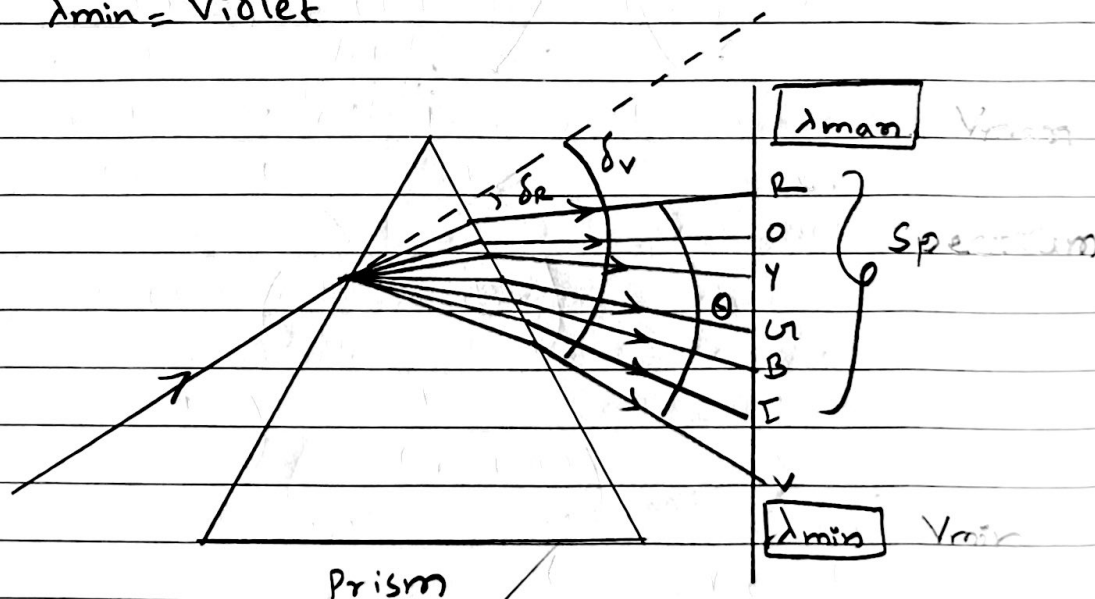
Dispersion of Light

→ The phenomenon of splitting white light into 7 distinct colors when passing from a prism is called dispersion of light.

The band of color obtained on the screen is called spectrum. The seven colours in the increasing order of wavelength are V, I, B, G, Y, O, R.

$\lambda_{\text{max}} = \text{Red}$

$\lambda_{\text{min}} = \text{Violet}$



• Angular dispersion (θ)

→ Angular dispersion is the Angle between two extreme colors (i.e. Red and Violet)

i.e. $\theta = \delta_V - \delta_R$

$$\Rightarrow \theta = A(\mu_v - 1) - A(\mu_r - 1)$$

$$\therefore \theta = A(\mu_v - \mu_r)$$

- Dispersive power (ω)

↳ It is the ability of an optical instrument to disperse the light.

It is defined as the ratio of angular dispersion to the deviation of mean light.

$$\text{i.e. } \omega = \frac{\theta}{\delta}$$

$$\Rightarrow \omega = \frac{A(\mu_v - \mu_r)}{A(\mu - 1)}$$

$$\therefore \omega = \frac{(\mu_v - \mu_r)}{(\mu - 1)}$$

It has no unit and is non-negative quantity.

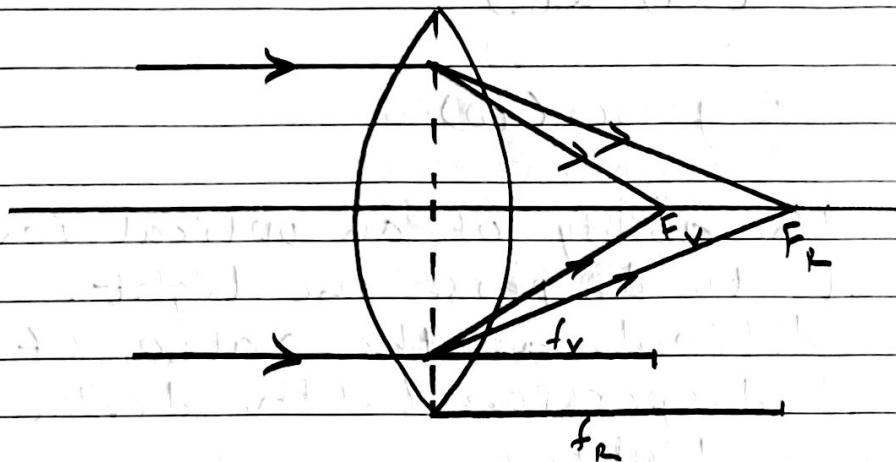
- Chromatic Aberration (Longitudinal axial).

↳ The inability of lens to focus all the seven colors at a point after refraction is called chromatic Aberration.

It is due to prismatic action of lens.

Mathematically,

$$F_R - F_V = \Delta f$$



Consider a thin convex lens of focal length f and Radii of curvature R_1 and R_2 . f_V and f_R be the focal length of two extreme colours. Now, From lens maker formula.

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\Rightarrow \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{1}{f(\mu - 1)} \rightarrow \text{①}$$

Using lens maker formula for red colour

$$\frac{1}{f_R} = (\mu_R - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f_R} = \frac{(\mu_R - 1)}{f(\mu - 1)} \rightarrow \text{②} \text{ Using eqn ①}$$

Similarly, for violet color,

$$\frac{1}{f_V} = \frac{(\mu_V - 1)}{f(\mu - 1)} \rightarrow \text{③}$$

subtracting eqn (ii) from (iii)

$$\frac{1}{f_v} - \frac{1}{f_r} = \frac{(u_v - 1)}{f(u_v - 1)} - \frac{(u_r - 1)}{f(u_r - 1)}$$

$$\Rightarrow \frac{f_r - f_v}{f_r f_v} = \frac{u_v - u_r}{f(u_v - 1)}$$

$$\Rightarrow f_r - f_v = \frac{(u_v - u_r)}{(u_v - 1)} \times \frac{f_r f_v}{f}$$

$$\Rightarrow f_r - f_v = \omega \frac{f^2}{f} \quad \therefore f = \sqrt{f_r f_v}$$

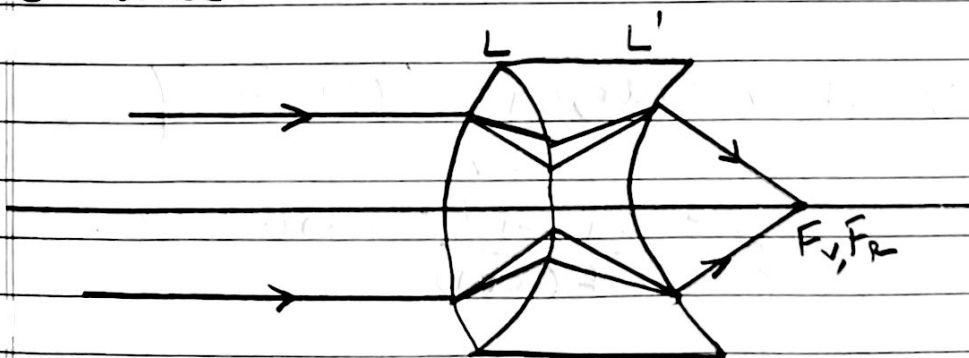
$$\boxed{\therefore f_r - f_v = \omega f}$$

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Achromatic combination of lens.

→ The combination of two lens which is free from chromatic ~~aberr~~ aberration is called Achromatic combination of lens.

For this, concave lens and convex lens of suitable focal length are combined.



Let us consider two lenses L and L' made up of different materials of dispersive powers ω and ω' respectively. Let μ_v, μ, μ_r and f_v, f, f_r be the refractive indices and focal lengths of violet, mean and red colours of light for lens L .

Let μ'_v, μ' and μ'_r and f'_v, f' and f'_r be the refractive indices and focal lengths of violet, mean and red colours of light of lens L' .

For lens L :

focal length of mean colour is :

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{f(\mu - 1)} \rightarrow (i)$$

For Red colour, $\frac{1}{f_r} = (\mu_r - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$

$$\frac{1}{f_r} = \frac{\mu_r - 1}{f(\mu - 1)} \rightarrow (ii)$$

For Violet colour, $\frac{1}{f_v} = (\mu_v - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$

$$\therefore \frac{1}{f_v} = \frac{\mu_v - 1}{f(\mu - 1)} \rightarrow (iii)$$

For lens L' :

focal length of mean colour:

$$\frac{1}{f'} = (\mu' - 1) \left(\frac{1}{R_1'} + \frac{1}{R_2'} \right)$$

$$\Rightarrow \frac{1}{f'(\mu' - 1)} = \frac{1}{R_1'} + \frac{1}{R_2'} \rightarrow (iv)$$

For Red colour, $\frac{1}{f'_r} = (\mu'_r - 1) \left(\frac{1}{R_1'} + \frac{1}{R_2'} \right)$

$$\Rightarrow \frac{1}{f'_r} = \frac{\mu'_r - 1}{f'(\mu' - 1)} \rightarrow (v)$$

For ~~red~~ Violet colour.

$$\frac{1}{f'_v} = \frac{\mu'_v - 1}{f'(\mu' - 1)} \rightarrow (vi)$$

For achromatism, combined focal lengths for violet colour must be equal to that for red colour.

$$F_v = F_r$$

$$\text{or, } \frac{1}{F_v} = \frac{1}{F_r}$$

$$\Rightarrow \frac{1}{f_v} + \frac{1}{f_v'} = \frac{1}{f_r} + \frac{1}{f_r'}$$

$$\Rightarrow \frac{1}{f_v} - \frac{1}{f_r} = \frac{1}{f_v'} - \frac{1}{f_r'}$$

$$\Rightarrow \frac{\mu_v - \mu_r}{f(\mu - 1)} = \frac{\mu_v' - \mu_r'}{f'(\mu' - 1)}$$

$$\Rightarrow \frac{\omega}{f} = -\frac{\omega'}{f'}$$

This is the required equation.

Since the dispersive powers are always positive, the above equation is possible only when f and f' are of opposite nature.

If $\omega = \omega'$, $\frac{1}{f} + \frac{1}{f'} = 0$, i.e. combination behaves as a glass slab.

Q. The focal length of Conver lens for red light and Violet light are 26 cm and 25 cm respectively. Calculate the dispersive power of material of lens. Also find chromatic aberration produced by lens.

→ Soln,

$$\text{Use formula } \omega = \frac{f_r - f_v}{f} = \frac{f_r - f_v}{\sqrt{f_r f_v}} = 0.04$$

$$\text{Here, chromatic aberration} = f_r - f_v = 1 \text{ cm}$$

Q. An achromatic converging lens of mean focal length 40 cm is made by combining two lenses of different materials. If the dispersive powers of the two lenses are in the ratio 1:3, find the focal length of each lens.

→ Solution,

$$\text{Here, } F = 40 \text{ cm}$$

The condition of achromatism is

$$\frac{1}{f'} = -\frac{w}{w'} = -\frac{1}{3}$$
$$f' = -3f$$

Again,

$$\frac{1}{F} = \frac{1}{f} + \frac{1}{f'} = \frac{1}{f} - \frac{1}{3f} = \frac{2}{3f}$$

on solving, $f = 26.67$, $f' = -80 \text{ cm}$.

• Pure and Impure Spectrum:

↳ The spectrum in which the constituent colours overlap each other and various colours are not clearly visible is called impure spectrum.
For example, rainbow.

↳ The spectrum in which there is no overlapping of colours and each colour is distinctly visible is called pure spectrum.

• Scattering of Light

↳ When light travels in medium, it interacts with atoms or molecules of the medium where it gets absorbed by them. These molecules then re-radiate the light in a direction away from the original direction of travel. The re-radiated light by ~~molecules~~ are of same colour.

This phenomenon of absorption and re-radiation of the light by molecules of medium is called scattering.

According to Lord Rayleigh, Scattering intensity of radiation is found to be inversely proportional to the fourth power of the wavelength.

$$\text{i.e. } I \propto \frac{1}{\lambda^4}$$

i.e. light of shorter wavelength scatters more light of larger wavelength.

- Blue Colour of sky: Since wavelength of blue colour shorter than red colour, the scattering of blue colour is more greater than red colour. So sky appears blue.

In fact, Violet is much more scattered ~~than~~ than blue colour, but our eye is more sensitive to blue than violet. Hence sky does not appear violet.

Conceptual (short) questions

① When white light is dispersed by a prism, red light appears at the top of the spectrum whereas violet at the bottom, why?

Ans: The ~~ref~~ refractive index (n) of glass is different for ~~diff~~ different colors. According to Cauchy's formula, refractive index is inversely proportional to wavelength. Since red light has the longest wavelength, it experiences the least refractive index and deviates the least. Violet has the shortest wavelength, experiences the highest refractive index, and deviates the most, placing it at the bottom.

② Why does the sky appear blue?

Ans: This is due to Rayleigh scattering. The intensity of scattered light is inversely proportional to the ~~power~~ fourth power of its wavelength.

Since blue light has a shorter wavelength than red, it is scattered much more strongly by the molecules in the atmosphere, filling the sky with blue light.

③ Draw an outline sketch to show the chromatic aberration in the case of a lens. How can achromatism be achieved?

Ans: Achromatism is achieved by combining two len-

ses usually a convex lens and concave lens such that their total dispersion is zero while maintaining some net focal length.

④ A spherical mirror cannot give rise to chromatic aberration. Why?

Ans Chromatic aberration is a result of refraction where the focal length depends on the refractive index of the material. ~~Reflection~~ Since mirrors work on the principle of reflection, which is independent of wavelength, all colors reflect at the same angle. Therefore, mirrors do not produce chromatic aberration.

⑤ The sun looks red at sun-rise and sun-set. Why?

Ans At these times, sunlight travels a longer distance through the atmosphere. Shorter wavelengths (blue, violet) are scattered away, while the longer wavelength (red) survives the journey to reach our eyes.

⑥ If a plane glass slab is placed on letter of different colors, then violet coloured letters appear more and more raised up. Why?

Ans The apparent shift (s) of an object under a glass slab is $s = t \left(1 - \frac{1}{\mu} \right)$. Since the

refractive index for violet light (μ_v) is greater than that for red light (μ_r), the apparent shift for violet is greater, making those letters appear more raised.

- ⑦ The eye is more sensitive to yellow color. But the danger signals are red, why?

Ans While the eye is technically most sensitive to yellowish-green (555 nm) in bright light, red is used for danger signals because it has the longest wavelength and is less scattered by fog, dust, or smoke, making it visible from the greatest distance.

- ⑧ An achromatic converging lens of mean focal length 40 cm is made by combining two lenses of different materials. If the dispersive powers of two lenses are in the ratio 1:3, find the focal length of each lens.

Ans Given,

$$F = 40 \text{ cm}$$

$$\frac{\omega_1}{\omega_2} = \frac{1}{3}$$

$$\text{or, } \omega_2 = 3\omega_1$$

Now,

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$

$$\text{or, } \frac{\omega_1}{f_1} = -\frac{3\omega_1}{f_2}$$

$$\text{or, } \frac{1}{f_1} = -\frac{3}{f_2}$$

$$\therefore f_2 = -3f_1$$

Now,

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\text{or, } \frac{1}{40} = \frac{1}{f_1} - \frac{1}{3f_1}$$

$$\text{or, } \frac{1}{40} = \frac{3-1}{3f_1}$$

$$\text{or, } 40 = \frac{3f_1}{2}$$

$$\therefore f_1 = 26.67 \text{ cm}$$

$$\therefore f_2 = -3f_1 = -3 \times \frac{80}{3} = -80 \text{ cm,}$$

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